

GARUDA

(Web application)

ADDITIONAL SECTION AI-BASED ANALYTICS & INTELLIGENCE FEATURES

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ADDITIONAL SECTION
AI-BASED ANALYTICS & INTELLIGENCE FEATURES
For Inclusion in DPR & SRS

AI-BASED ANALYTICS MODULE

1. Introduction:

To strengthen intelligence-led anti-narcotics enforcement, the proposed web application shall incorporate Artificial Intelligence (AI), Machine Learning (ML), and Advanced Data Analytics for predictive policing, offender behavior analysis, network detection, hotspot identification, and automated intelligence generation.

The AI module shall assist investigating officers, intelligence units, narcotics enforcement teams, and district police in identifying hidden criminal patterns, repeat offender trends, interstate drug trafficking networks, and suspicious operational activities.

2. Objectives of AI-Based Analytics

The AI analytics engine aims to:

- Predict emerging narcotics hotspots.
- Identify repeat offender patterns.
- Detect organized drug trafficking networks.
- Analyze movement and communication patterns.
- Identify hidden associations between accused persons.
- Generate automated intelligence alerts.
- Improve preventive policing and surveillance.
- Support strategic anti-drug enforcement planning.

3. AI Analytics Functional Components

3.1 Predictive Crime Analytics

a. Features

- Predict future narcotics hotspots.
- Analyze seasonal and event-based drug movement patterns.
- Predict high-risk areas.
- Forecast repeat offender activity.

b. Inputs

- Historical NDPS case data
- GPS locations
- Time-based trends
- Arrest history
- Intelligence reports

c.Outputs

- Risk heat maps
- High-risk zone alerts
- Preventive deployment recommendations

3.2 Repeat Offender Risk Scoring

a.AI Risk Parameters

- Number of previous NDPS cases
- Bail history
- Interstate links
- Association with organized networks
- Financial activity patterns
- Frequency of police verification

b.Output

Each accused shall be assigned:

- Low Risk
- Medium Risk
- High Risk
- Organized Network Risk

3.3 Criminal Network & Link Analysis

a. AI Functions

- Identify hidden associations.
- Detect supplier-peddler-consumer hierarchy.
- Generate network diagrams automatically.
- Identify key network operators.

b.Linkage Parameters

- Mobile numbers
- IMEI numbers
- Vehicles
- Addresses
- Bank accounts
- Associates
- Social media links

c.Outputs

- Network visualization
- Kingpin identification
- Common associates analysis

3.4 Geo-Spatial AI Analytics

a.Features

- AI-based hotspot mapping
- Route prediction
- Drug corridor analysis
- Interstate transport route detection

b.Outputs

- Heat maps
- Risk corridors
- Vulnerable village/area identification

3.5 Behavioral Pattern Analysis

a.AI Analysis

- Movement patterns
- Communication frequency
- Repeated travel routes
- Time-based activity trends

b.Outputs

- Suspicious movement alerts
- Night movement analysis
- Recurring operational patterns

3.6 Facial Recognition Integration (Future Scope)

a.Features

- Accused face matching
- CCTV integration
- Surveillance identification
- Missing/wanted accused detection

b.Integration Sources

- CCTV feeds
- Police databases
- Prison databases

3.7 Vehicle Intelligence Analytics

a.Features

- Frequently used vehicle detection
- Suspicious vehicle movement analysis
- Interstate vehicle tracking
- Repeat vehicle association analysis

b.Outputs

- High-risk vehicle alerts
- Drug transport vehicle patterns

3.8 Mobile & Communication Intelligence

a.Features

- Common mobile number detection
- SIM swapping analysis
- Frequent contact analysis
- Tower location clustering

b.Outputs

- Communication network mapping
- Common handler identification

3.9 AI-Based Intelligence Alerts

a. Automated Alerts

- Repeat offender entering hotspot area
- Bail release alerts
- Interstate movement alerts
- Frequent association alerts
- Sudden increase in NDPS activity
- Suspicious communication clusters

3.10 Social Media Monitoring Analytics

a. Features

- Drug-related keyword monitoring
- Suspicious social media profiles
- Encrypted communication indicators
- Dark web intelligence inputs (future scope)

4. AI Dashboard Features

a. Dashboard Components

Real-Time Widgets

- Active hotspots
- High-risk accused
- Interstate movement alerts
- Network alerts
- Surveillance due list

b. Graphical Analytics

- Crime trend graphs
- Heat maps
- Association charts
- Drug category analysis

5. Machine Learning Capabilities

ML Functions

- Pattern recognition
- Risk prediction
- Anomaly detection
- Behavioral learning
- Automated classification

6. AI Data Sources

The AI engine shall analyze:

- FIR data
- Arrest records
- Mobile data
- Vehicle data
- Surveillance reports
- Intelligence inputs
- GIS coordinates

- CCTV metadata
- Criminal history sheets

7. AI Risk Categorization System

Sl.No.	Risk level	Description
1	Low	First-time offender
2	Medium	Repeat involvement
3	High	Habitual trafficker
4	Critical	Organized syndicate operator

8. AI-Based Predictive Enforcement

The system shall recommend:

- Surprise checks
- CASO operations
- Surveillance strengthening
- Targeted vehicle checks
- Lodge verification
- Beat officer verification

9. Security & Ethical Controls for AI

Controls

- Human approval for sensitive actions
- AI audit logs
- Access control
- Data privacy protection
- Bias prevention mechanisms
- Controlled intelligence sharing

10. Infrastructure Requirements for AI

a. Hardware

- GPU-enabled servers
- High-performance storage
- AI processing engine

b. Software

- AI analytics engine
- Machine learning framework
- GIS analytics engine

11. Future AI Enhancements

Proposed Future Features

- Drone analytics
- Crowd intelligence
- Voice pattern analysis
- Predictive syndicate detection
- AI-powered cyber narcotics monitoring

SRS ADDITION — AI ANALYTICS MODULE

- Functional Requirements — AI Module
 - FR-AI-1 Predictive Analytics
 - Description
- a. System shall predict probable NDPS hotspots using historical data.
- 1. Inputs
 - Historical FIR data
 - GIS data
 - 2. Outputs
 - Heat maps
 - Risk alerts
 - 3. FR-AI-2 Risk Scoring Engine
 - 4. Description
- b. System shall assign AI-generated risk scores to accused persons.
- FR-AI-3 Network Analysis
- 1. Description
- c. System shall generate automated association networks.
- 1. FR-AI-4 AI Alerts
 - 2. Description
- c. System shall generate automated intelligence alerts.
- 1. FR-AI-5 Facial Recognition
 - 2. Description
- d. System shall support accused face matching using CCTV footage.
- 1. Non-Functional Requirements — AI Module
 - 2. Performance
 - 3. AI response within defined limits.
 - 4. Scalability
 - Support statewide AI analytics.
- e. Security
- AI-generated data encryption.
- f. Accuracy
- Continuous model training and validation.

AI REPORTS

Proposed Reports

- High-risk accused report
- Hotspot analysis report
- Repeat offender report
- Interstate trafficking report
- Network association report
- AI risk scoring report